

Jingbin (Justin) Lin

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EDUCATION

University of California - San Diego

Master of Science, Computer Engineering

Sep 2025 - Present

California

- **Coursework:** Low-power VLSI Implementation for Machine Learning, Digital Communication Circuits I, Microwave Circuits, VLSI Digit Sys Algorithms & Arch, VLSI Integrated Circuits and Systems Design, Advace Computer Arch, System Design for Machine Intelligence

McMaster University

Bachelor of Engineering, Electrical Engineering and Management (GPA: 3.5)

Sep 2020 - Jun 2025

Canada

- **Achievements:** Dean's Honour List, McMaster Honour Award, Dr. Rudolf De Buda Scholarship
- **Coursework:** Microelectronic (CMOS Technology), Photonic Device & Systems, Digital Communication System, Electric Motor Drives

SKILLS

- **Languages:** Verilog, C++, Python, TypeScript
- **Tools/Library:** GitHub, Jira, Pytorch, LangGraph, VSCode
- **Devices:** Linux

AWARDS

- **Dean's Honour List:**(2025, 2023, 2021)
- **Dr. Rudolf De Buda Scholarship:**(2025)
- **McMaster Honour Award:**(2020)

PROJECTS

Consent-Based LLM-OSINT Agent System | [Project Link](#)

Jan 2026 - Mar 2026

- Built an end-to-end LLM-powered OSINT pipeline that generates structured, citation-grounded person profiles from public data using an agentic workflow (LangGraph + MCP tools)
- Designed hybrid retrieval architecture (vector index + knowledge graph) with provenance tracking to improve consistency, auditability, and hallucination mitigation
- Implemented graph cleanup and normalization pipeline (entity resolution, deduplication, conflict arbitration) to reduce contradictions across multi-source data
- Evaluated on multiple consenting subjects, producing 20+ page structured reports and demonstrating improved coverage and reliability in automated profiling

2-D Systolic Array Design with Convolution Layer Implementation

Sep 2025 - Dec 2025

[Project Link](#)

- Design 2D Systolic Array and train CNN model (VGG and ResNet-20) to compute convolution.
- Simulation test on FPGA Cyclone IV GX with performance metric: power consumption, through put, operation frequency, etc.
- Trained VGGNet on PyTorch based on CIFAR10 Dataset, modify layer size and prune using C2F method to save compute power.
- Pruned the CNN to 88% sparsity while maintaining >90% validation precision for both architecture

Pi Cluster for Parallel Computing | [Project Link](#)

May 2025 - Jul 2025

- Builded a Raspberry Pi cluster with 5 nodes and compact size for destributed computing environment simulation
- Setuped OpenMPI as computing library, DHCP+SLURM for cluster interconnection and management
- Added a real-time, single direction monitor page with gist
- Successfully ran OpenMPI jobs in python and C++, with up to 16 cores distributed in 4 nodes

Autonomous Electric Vehicle Simulation | [Project Link](#)

Sep 2024 - Dec 2024

- Constructed AEV using Nvidia Jetson Nano while applying ROS2 system, Depth Camera, LiDAR and IMU sensor
- Solved mapping and route planning issues by employing SLAM, Quadratic Propagation and odometry transformation algorithm
- Successfully ran vehicle through multiple scenarios with 60fps SLAM

Wearable Cave Mapping System | [Project Link](#)

Jan 2025 - Apr 2025

- Designed an affordable, wearable 3D scanning device for cave scanning.
- Lead a team of 5, integrated multiple devices like IMU, LiDAR, Depth Camera for localization, position tracking and mapping
- Achieved 3 hour work time with 15fps mapping update

WORK EXPERIENCE

MediumAI | Co-Founder&Software Engineer

Jan 2023 - Dec 2025

- Built production RAG LLM pipelines for multilingual medical transcription and multimodal documentation using Langgraph, vector database, and cloud architecture, improving transcription accuracy in Mandarin and Cantonese by 10%, reducing time to first token by 32%
- Led R&D planning and delivery using GitHub and Jira, coordinated a team of engineers to cut operating costs by 42% through a multi-agent, multi-modal approach
- Manage the R&D team and plan work using GitHub and Jira, coordinating tasks and tracking progress to keep projects on schedule and ensure timely delivery of milestones
- Secured \$10,000+ investment at the initial stage through a technical demo and assisted expansion to over 15 clients in Toronto, enabling early market traction

Sun Yat-sen University | Computer Science RA Intern

May 2023 - Aug 2023

- Developed a neural network for physics simulation using Python, enabling more accurate modeling of physical processes
- Researched and tested MLP neural networks with NVIDIA Modulus as a differential equation solver, improving solution accuracy for complex equations
- Used Open3D and Paraview to conduct post-processing on dataset and visualize computing result
- Developed a dataset conversion algorithm in Python that transformed 3D scans STL files into pointcloud dataset, simplifying data preparation for simulation tasks